

INTRODUCTION

This document outlines the entity-relationship (ER) diagram for the mini-world we described as part of Phase 1 of this project. What follows is a design quality analysis of the ER diagram. Please refer to Figure 1 for the diagram itself.

There are a few small changes we have made on our model from the first phase:

1. Participation constraint of Ghoul and Human in the relationship GhoulPart and HumanPart have been updated to be partial instead of total. This is crucial as not all ghouls and humans need to take part in the relationship.
2. Cardinality ratio of Human and Ward participating in the relationships Victim and Ward have been updated to 1. This replicates the real-world scenario appropriately.
3. To uphold the ER-diagram readability (Left-to-Right and Up-to-Down flow), we have renamed the relationship LivesIn to InhabitedBy and MadeFrom to UsedIn.
4. Updated the time attribute of the weak entities Encounter and GhoulMurder to be partial keys. This helps uniquely identify the instances when coupled with the primary keys of the entity in their respective identifying relationships. The OneEyedGhoul weak-entity does not need a partial key since all its identifying relationships are one-to-one.

DESIGN QUALITY ANALYSIS

Redundancy Analysis

For one-eyed ghouls, some data is duplicated between the human-part and ghoul-part. Most of this data will be different (e.g. the ghoul could have different names in human and ghoul societies), but some times the data may be the same (e.g. gender). This doesn't cause any major problems since the one-eyed ghoul can live different human and ghoul lives. Sometimes this redundancy becomes useful, in cases where the CCG wants to track both lives of the ghoul.

Update Anomaly

When a ghoul/human is killed by an encounter/murder, we have to make sure to set their respective deceased attributes to true. This is an update anomaly.

Insertion Anomaly

There is no insertion anomaly in our model because all our relationships are atomic, concrete, and coherent.

Deletion Anomaly

There is no deletion anomaly in our model. In fact, there isn't much opportunity for deleting rows (apart from mis-insertions) since this database largely serves a record-keeping purpose.

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